Problem Statement

Evaluate the limit from the image "dailyintegral-limit-limit-1x1-2026-07-31.jpg":

$$L = \lim_{n\to\infty} n\left[\left(1+\frac{3}{n}\right)^n - e^3\right]$$

Detailed Solution

To evaluate the limit L, we need to find an asymptotic expansion for the term $\left(1+\frac{3}{n}\right)^n$ as $n \to \infty$.

Step 1: Rewrite using Exponential and Logarithm

We can rewrite the expression $\left(1+\frac{3}{n}\right)^n$ using the base e:

$$\left(1+\frac{3}{n}\right)^n = \exp\left[n\ln\left(1+\frac{3}{n}\right)\right]$$

Step 2: Taylor Expansion of the Logarithm

Recall the Taylor series expansion for $\ln(1+x)$ when x is near 0:

$$\ln(1+x) = x - \frac{x^2}{2} + \mathcal{O}(x^3)$$

Substituting $x = \frac{3}{n}$, which approaches 0 as $n \to \infty$:

$$\ln\left(1+\frac{3}{n}\right) = \frac{3}{n} - \frac{1}{2}\left(\frac{3}{n}\right)^2 + \mathcal{O}\left(\frac{1}{n^3}\right) = \frac{3}{n} - \frac{9}{2n^2} + \mathcal{O}\left(\frac{1}{n^3}\right)$$

Step 3: Evaluate the Exponent

Now, multiply the expansion by n:

$$n\ln\left(1+\frac{3}{n}\right) = n\left(\frac{3}{n} - \frac{9}{2n^2} + \mathcal{O}\left(\frac{1}{n^3}\right)\right) = 3 - \frac{9}{2n} + \mathcal{O}\left(\frac{1}{n^2}\right)$$

Substituting this back into the exponential expression:

$$\exp\left[n\ln\left(1+\frac{3}{n}\right)\right] = \exp\left(3 - \frac{9}{2n} + \mathcal{O}\left(\frac{1}{n^2}\right)\right)$$

Using the property of exponents, $e^{a+b} = e^a e^b$:

$$\left(1+\frac{3}{n}\right)^n = e^3 \cdot \exp\left(-\frac{9}{2n} + \mathcal{O}\left(\frac{1}{n^2}\right)\right)$$

Step 4: Taylor Expansion of the Exponential

Use the Taylor series expansion $e^y = 1 + y + \mathcal{O}(y^2)$ for small y, with $y = -\frac{9}{2n} + \mathcal{O}\left(\frac{1}{n^2}\right)$:

$$\exp\left(-\frac{9}{2n} + \mathcal{O}\left(\frac{1}{n^2}\right)\right) = 1 - \frac{9}{2n} + \mathcal{O}\left(\frac{1}{n^2}\right)$$

Thus, the full term expands as:

$$\left(1 + \frac{3}{n}\right)^n = e^3\left(1 - \frac{9}{2n} + \mathcal{O}\left(\frac{1}{n^2}\right)\right) = e^3 - \frac{9e^3}{2n} + \mathcal{O}\left(\frac{1}{n^2}\right)$$

Step 5: Evaluating the Limit

Substitute this expansion back into the original limit expression for L:

$$L = \lim_{n\to\infty} n\left[\left(e^3 - \frac{9e^3}{2n} + \mathcal{O}\left(\frac{1}{n^2}\right)\right) - e^3\right]$$

The e^3 terms cancel out:

$$L = \lim_{n\to\infty} n\left[-\frac{9e^3}{2n} + \mathcal{O}\left(\frac{1}{n^2}\right)\right]$$

Distribute the n:

$$L = \lim_{n\to\infty}\left[-\frac{9e^3}{2} + \mathcal{O}\left(\frac{1}{n}\right)\right]$$

As $n \to \infty$, the $\mathcal{O}\left(\frac{1}{n}\right)$ term approaches 0.

$$L = -\frac{9e^3}{2}$$

Final Answer:

$$-\frac{\mathbf{9e^3}}{\mathbf{2}}$$